Paolo **Rabino**

paolotron | +39 327 731 2804 | paolo.srabino@gmail.com | paolo-rabino-072644201 | 0009-0001-2843-0614 | Google Scholar | Munich, Germany

Research Lead building learned 3D perception and motion generation for industrial robots. PhD in Artificial Intelligence (expected 2026), 6 publications, 1 patent, 100k+ GPU hours of competitive HPC grants.

Research Highlights

- **FoldPath** (IROS 2025): first end-to-end model generating long-horizon robot paths as neural fields; surpassed the state of the art on robotic spray painting, led to a **patent**
- **MultiGraspNet** (RA-L 2026): one multitask 3D model for parallel-jaw and vacuum grasping that **raised grasp success in real industrial trials** at lower computational cost
- **D3G** (RA-L 2025): end-to-end transformer for grasp-order reasoning in dense clutter; its metric was adopted by the **UNOBench** benchmark and the **ECCV 2026 challenge**; code and benchmark open-sourced (21★ on GitHub)
- **Efficient Odd-One-Out** (ICIAP 2025): state-of-the-art accuracy with **≈1/3 fewer parameters** and **≈3× faster training**; grew out of a co-supervised MSc thesis

Professional Experience

mkind — *Munich*
Research Lead — *2026 - Present*

- Lead research on **Physical AI** for **surface finishing** in high-mix industrial manufacturing
- Develop **trajectory generation** models building on FoldPath, and learn paint deposition from real 3D recordings of expert operators
- Lead applicant for **EuroHPC AI Factory** GPU allocations, including a granted 50,000 GPU-hour project

Robot Learning · 3D Vision · PyTorch · HPC

Politecnico di Torino — *Torino*
Teaching Assistant & Lab Instructor — *2022 - 2026*

- MSc courses: **Advanced Machine Learning**, **Machine Learning and Deep Learning**, **Data Analysis and Artificial Intelligence**
- BSc course **Algorithms and Data Structures** (C) and **Models for AI** (2nd-level Master in AI)
- Co-supervised **4 MSc theses** on anomaly detection, multimodal LLMs, point cloud registration and gripper selection, one leading to an ICIAP paper

Python · C · Machine Learning · Deep Learning

Motionness.ai / AnyPath — *Torino*
Co-Founder — *2025 - 2026*

- Co-founded an **AI** startup for **robotic path planning** from point clouds, later continued as AnyPath
- Built the full **MLOps** pipeline and full-stack product for learned robotic path generation

MLOps · Python · TypeScript · Full-Stack

Education

Politecnico di Torino & IIT, VANDAL Lab — *Italy*
National **PhD** in Artificial Intelligence (38th cycle) — *2022 - 2026 (expected)*

- Thesis: **From 3D Understanding to Action: Learning Methods for Industrial Robotics**, supervised by Prof. Tatiana Tommasi and co-supervised by Prof. Giuseppe Bruno Averta
- Topics: **3D Computer Vision** | **Robotic Manipulation** | **Anomaly Detection** | **Large Pre-Trained Models**
- Industry collaborations on robotic spray painting and multi-gripper bin picking; collaborator in the European BURG Project
- Reviewer for conferences **CVPR**, **ICCV**, **ECCV**, **AAAI**, **ICRA**, **IROS**, **ICIAP** and journals **TPAMI**, **T-RO**, **RA-L**, **IJRR**
- Attended the ICVSS 2023 International Computer Vision Summer School and the PoliTo PhD AI Fall School

Politecnico di Torino — *Italy*
Master of Science in Computer Engineering — *2021 - 2022*

- Thesis: Deep Learning for Robotic Grasping | **Graduated Cum Laude**

Politecnico di Torino — *Italy*
Bachelor of Science in Computer Engineering — *2019 - 2021*

Publications

Ortuno Chanelo, S., **Rabino, P.**, Civitelli, E., Tommasi, T., & Camoriano, R. (2026). MultiGraspNet: A Multitask 3D Vision Model for Multi-Gripper Robotic Grasping. *IEEE Robotics and Automation Letters (RA-L)*; poster at the Industrial Applications of Robot Learning Workshop @ IROS 2026. [arXiv]

Chito, S.*, **Rabino, P.***, Camoriano, R., & Tommasi, T. (2026). QuadricViT: Efficient 3D Semantic Occupancy Prediction. *Under submission*. (*equal contribution)

Rabino, P., Tiboni, G., & Tommasi, T. (2025). FoldPath: End-to-End Object-Centric Motion Generation via Modulated Implicit Paths. *IEEE/RSJ IROS*. [arXiv]

Chito, S., **Rabino, P.**, & Tommasi, T. (2025). Efficient Odd-One-Out Anomaly Detection. *ICIAP*. [arXiv]

Rabino, P. & Tommasi, T. (2025). A Modern Take on Visual Relationship Reasoning for Grasp Planning. *IEEE Robotics and Automation Letters (RA-L)*, presented at IROS. [arXiv]

Rabino, P., Alliegro, A., & Tommasi, T. (2024). 3D Semantic Novelty Detection via Large-Scale Pre-Trained Models. *IEEE Access*, 12, 135352–135361. [DOI]

Rabino, P., Alliegro, A., & Tommasi, T. (2024). Harnessing Large-Scale Pre-Trained Models for 3D Semantic Novelty Detection. *ECCV Workshops*.

Patents

Rabino, P., Tiboni, G., & Tommasi, T. (2025). Metodo per imparare la generazione del percorso motorio di robot / Robotic Path Generator from Point Clouds. *National and international patent application*, Politecnico di Torino. [Knowledge Share]

Honors, Awards & Grants

2026	**Best Poster Presentation Award, for MultiGraspNet**, 1st Int. Workshop on Industrial Applications of Robot Learning @ IEEE/RSJ IROS 2026	*Pittsburgh, USA*
2026	**Meta Project Aria research grant**, wrote the winning proposal to collect an egocentric dataset of human interactions with deformable objects; awarded Meta Project Aria glasses	
2023 - 2026	**Competitive HPC compute grants, over 100,000 GPU hours overall**, PI of 3 projects (2 EuroHPC AI Factory, 1 CINECA ISCRA) and collaborator on 5 CINECA ISCRA projects, on CINECA Leonardo and BSC MareNostrum 5	
2021 - 2022	**Board of Directors, IEEE-HKN honor society**, selected for academic performance; co-managed a non-profit with > 50 members and organized conferences and hackathons	*Torino*

Selected Projects

QuadricViT *CUDA C++, PyTorch C++/CUDA extensions, ViT*

Efficient 3D semantic occupancy prediction for autonomous driving (under submission). Wrote the **custom CUDA kernels** that aggregate thousands of superquadric primitives into the voxel occupancy grid, reformulating them in **log space** to remove the numerical instabilities of the original implementation. The model runs with ≈47% lower latency and ≈54% less memory than QuadricFormer on nuScenes.

baylib [GitHub] *CUDA C++, OpenCL, C++20, multithreading*

Co-developer (second-largest contributor) of a high-performance C++ library for approximate inference on discrete Bayesian networks. Wrote the **GPU backend**: **custom CUDA kernels** for logic sampling and likelihood weighting (with evidence support, kernel optimizations and platform-independent builds), and **OpenCL** implementations of logic and adaptive importance sampling with a CPU fallback.

BlokusRL [GitHub] *Python, reinforcement learning, W&B*

Custom simulator and environment for the board game Blokus, used to train reinforcement-learning agents.

Skills

Languages	English (professional proficiency) | Italian (native)
Programming	Python | C++/CUDA | C | TypeScript/JavaScript | SQL | Bash | LaTeX
GPU Computing	CUDA C++ (custom kernels) | PyTorch C++/CUDA extensions | OpenCL / Boost.Compute | C++ multithreading | Multi-GPU training on HPC clusters (CINECA Leonardo, BSC MareNostrum 5)
Deep Learning	PyTorch | PyTorch Geometric | PEFT | Accelerate | vLLM | ONNX | FAISS | scikit-learn | W&B
3D & Vision	PyTorch3D | Open3D | Trimesh | Mask2Former | FoundationPose | DINOv2
Infrastructure	MLOps | Git
Tools	Blender | Inkscape | Adobe Illustrator | VS Code | JetBrains | Vim
Interests	Reading | Board Games | Hiking